Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you have a question, raise your hand and I will come to you.
- You have 50 minutes to complete the exam.
- Good luck!

Question:	1	2	3	4	5	Total
Points:	10	10	10	10	10	50
Score:						

1. (10 points) For each of the following statements, write the negation of the statement without using the logical operator "$\neg$" or the word "not." Determine whether the original statement or its negation is true, and give a proof of the true statement.

   (a) There exists $a \in \mathbb{N}$ such that, for all $b \in \mathbb{N}$, if ab is even then $a + b$ is odd.

In symbols, this says

$$(\exists a \in \mathbb{N})(\forall b \in \mathbb{N})(ab \text{ is even} \longrightarrow a+b \text{ odd})$$

The negation is

$$(\forall a \in \mathbb{N})(\exists b \in \mathbb{N})(ab \text{ is even and } a+b \text{ is even})$$

The original statement is True. In fact, Take $a = 1 \in \mathbb{N}$. Given any $b \in \mathbb{N}$, if ab is even, since $ab = 1b = b$, we know that b is even, say $b = 2k$ with $k \in \mathbb{Z}$.
Hence $a+b = 1+2k$ is odd, as desired.

   (b) For all $x, y \in \mathbb{R}$, if $x + y > 5$ then either $x > 0$ or $y > 0$.

In symbols, this says:

$$(\forall x, y \in \mathbb{R})(x+y>5 \longrightarrow (x>0 \text{ or } y>0)$$

The negation is

$$(\exists x, y \in \mathbb{R})(x+y>5 \text{ and both } x \leq 0 \text{ and } y \leq 0)$$

The original statement is True. Given $x, y \in \mathbb{R}$, we prove the contrapositive: if $x \leq 0$ and $y \leq 0$, then $x+y \leq 5$

If $x \leq 0$ and $y \leq 0$, then $-x \geq 0$ and $-y \geq 0$, so $-x-y \geq 0$ by positive closure. But $-x-y = (-1)x + (-1)y = (-1)(x+y) = -(x+y)$. Since $-x-y \geq 0$, its additive inverse, which is $x+y$, is ≤ 0, so also ≤ 5.

2. (10 points) Let P and Q be statements.

   (a) Fill in the truth table below with the appropriate truth values.

P	Q	$\neg P$	$\neg Q$	$P \wedge \neg Q$	$\neg P \vee Q$	$\neg Q \to (P \wedge \neg Q)$	$(\neg P \vee Q) \to Q$
T	T	F	F	F	T	T	T
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	F	F

   (b) How are the statements $\neg Q \to (P \wedge \neg Q)$ and $(\neg P \vee Q) \to Q$ related? Circle all that apply (you don't need to justify this)

   (i) They are converses of each other.

   (ii) They are contrapositives of each other.

   (iii) They are converses of each other.

   (iv) They are logically equivalent.

3. (10 points) Decide whether the following statements are true or false. If the statement is true, prove it. If it is false, give a counterexample.

   (a) For any sets A, B and C, we have $(A \setminus B) \cap C = (A \setminus C) \cup (A \setminus B)$.

This is False. Let $A = \{1\}$, $B = \{1\}$, $C = \{2\}$

Then $(A \setminus B) \cap C = \emptyset \cap C = \emptyset$, but

$(A \setminus C) \cup (A \setminus B) = \{1\} \cup \emptyset = \{1\}$

So $(A \setminus B) \cap C \neq (A \setminus C) \cup (A \setminus B)$

   (b) For every $x, y, z \in \mathbb{R}$ with $z \neq 0$, if $x < y$ then $\frac{x}{z} < \frac{y}{z}$.

This is False. We show a counterexample.

Let $x = 1$, $y = 2$ and $z = -1$. Then $z \neq 0$ and $x < y$,

but $\frac{x}{z} = -x = -1$ and $\frac{y}{z} = -y = -2$, so $\frac{x}{z} > \frac{y}{z}$.

4. (10 points) Prove that, for every natural number n, we have $\sum_{i=1}^{2n}(-1)^i i = n$.

Proof: We proceed by induction on n.

Base case: For $n=1$, $\sum_{i=1}^{2n}(-1)^i i = \sum_{i=1}^{2}(-1)^i i = -1+2 = 1 = n$

So the base case holds.

Inductive Step: Fix $k \in \mathbb{N}$ and assume that $\sum_{i=1}^{2k}(-1)^i i = k$.

We want to show that $\sum_{i=1}^{2(k+1)}(-1)^i i = k+1$.

We compute, starting with the left-hand side:

$$\sum_{i=1}^{2(k+1)}(-1)^i i = \sum_{i=1}^{2k+2}(-1)^i i = \boxed{\sum_{i=1}^{2k}(-1)^i} + \underbrace{(-1)^{2k+1}(2k+1)}_{i=2k+1} + \underbrace{(-1)^{2k+2}(2k+2)}_{i=2k+2}$$

Ind Hyp

$$= \boxed{k} - (2k+1) + 2k+2$$

$$= k - 2k - 1 + 2k + 2$$

$$= k+1, \quad \text{as desired.}$$

(We used that $(-1)^{2k+1} = -1$ because $2k+1$ is odd and that $(-1)^{2k+2} = 1$ because $2k+2 = 2(k+1)$ is even)

It follows by induction that

$$\sum_{i=1}^{2n}(-1)^i i = n \quad \text{for every natural } n \qquad \square$$

5. (10 points) Prove that, for each $n \in \mathbb{N} \cup \{0\}$, there are F_n (the n-th Fibonacci number) different ways to tile a $1 \times n$ grid using squares (i.e. 1x1 tiles) and dominoes (i.e. 1x2 tiles). Different here means that if we look from left to right a different sequence of tiles has been used. Assume that there is 1 way to tile a 1×0 grid.

   Recall that the Fibonacci numbers are defined by $F_0 = 1$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geqslant 2$.

Proof: By strong induction on $n \geq 0$.

Base cases: For $n=0$, we know there is 1 way to tile a 1×0 grid, and also $F_0 = 1$

For $n=1$ there is 1 way to tile a 1×1 grid, and $F_1 = 1$.

Inductive step: Assume that, for some $K \geq 1$, there are F_j ways to tile a $1\times j$ grid for every $1 \leq j \leq K$.

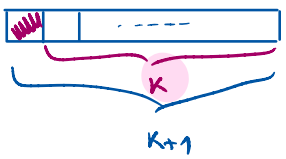
Here we can assume $K \geq 1$ instead of $K \geq 0$ because we treated $n=1$ in the base case.

Take a $1 \times (K+1)$ grid. We want to show it can be tiled in exactly F_{K+1} different ways.

In any tiling, the left-most square is covered either by a 1×1 square or a 1×2 domino, but not both.

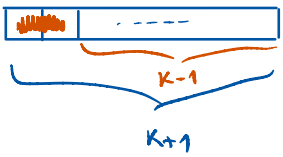
Hence the total number of tilings can be computed by adding these two possibilities

Case 1: Left-most square is covered by a 1×1 square:



In this case, what remains is a $1\times K$ grid, so we know by ind. hyp. that it can be tiled in exactly F_K different ways

Case 2: Left-most square is covered by a 1×2 square.



In this case, what remains is a $1\times(K-1)$ grid, which can be tiled in exactly F_{K-1} different ways by inductive hyp.

Hence, a $1\times(K+1)$ grid can be tiled in $F_K + F_{K-1}$ different ways

But, since $K \geq 1$, we have $F_{K+1} = F_K + F_{K-1}$ by definition

So the statement is also true for $K+1$.

By induction, the statement is true for every $n \geq 0$ □

Axioms

Field axioms

1. **Closure:** If $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
2. **Commutativity:** If $a, b \in \mathbb{R}$, then $a + b = b + a$ and $ab = ba$.
3. **Associativity:** If $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
4. **Distributivity:** If $a, b, c \in \mathbb{R}$, then $(a+b)c = ac+bc$ and $a(b + c) = ab + ac$.
5. **Existence of Identities:** There are two different real numbers 0 and 1 such that for all real numbers a, $a + 0 = a$ and $a \cdot 1 = a$.
6. **Additive Inverses:** For all $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
7. **Multiplicative Inverses:** For all $a \in \mathbb{R}$ such that $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

- $\mathbb{N} \cup \{0\}$ satisfies Axioms 1–5 (quantified over $\mathbb{N}$)
- $\mathbb{Z}$ satisfies Axioms 1–6 (quantified over $\mathbb{Z}$)
- $\mathbb{Q}$ satisfies Axioms 1–7 (quantified over $\mathbb{Q}$)

Equality Axioms. These hold for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

8. **Reflexivity:** For every real number a, $a = a$.
9. **Symmetry:** For all real numbers a and b, if $a = b$, then $b = a$.
10. **Transitivity:** For all real numbers a, b, and c, if $a = b$ and $b = c$ then $a = c$.
11. **Substitution:** If $a = b$, then b may be substituted for a in any statement.

Order Axioms. We assume $\mathbb{R}$ contains a subset $\mathbb{R}^+$ of "positive real numbers" such that

12. **Positive Closure:** If $a, b \in \mathbb{R}^+$, then $a + b \in \mathbb{R}^+$ and $ab \in \mathbb{R}^+$.
13. **Trichotomy:** If $a \in \mathbb{R}$, then exactly one of the following is true:

    • $a \in \mathbb{R}^+$ • $-a \in \mathbb{R}^+$ • $a = 0$.

14. **Well-Ordering Property:** If S is a nonempty set of positive integers, then S has a minimum element.

Facts

Logic. Let P, Q and R be statements.

- $P \to Q \equiv \neg P \vee Q$
- $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$

De Morgan's Laws

- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Distributive Laws

- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

Sets. Let A and B be subsets of universe U.

- $A \subseteq B$ if $x \in A$ implies $x \in B$.
- $A = B$ if $A \subseteq B$ and $B \subseteq A$.
- $A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$
- $A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$
- $A \backslash B = \{x \in U : x \in A \text{ and } x \notin B\}$
- $A^c = \{x \in U : x \notin A\}$

Inequalities. Let $a, b \in \mathbb{R}$.

- We say $a < b$ if $b - a \in \mathbb{R}^+$.
- We say $a \leqslant b$ if $a < b$ or $a = b$.
- We say $a > b$ if $a - b \in \mathbb{R}^+$.
- We say $a \geqslant b$ if $a > b$ or $a = b$.

Real numbers. Let $a, b, c \in \mathbb{R}$.

R1 If $a = b$ then $a + c = b + c$ and $ac = bc$.
R2 If $a + c = b + c$ then $a = b$.
R3 The additive identity in $\mathbb{R}$ is unique.
R4 $0 \cdot a = 0$.
R5 $(-1)a = -a$.
R6 $(-1)(-1) = 1$
R7 $(-a)(-b) = ab$
R8 $-(ab) = (-a)b = a(-b)$.
R9 1 is positive.
R10 $a \in \mathbb{R}^+$ if and only if $a > 0$.
R11 if $a > b$ and $c > 0$ then $ac > bc$.
R12 if $a > b$ then $a + c > b + c$.